

Task 2 Report

Task 2: Exploratory Data Analysis (EDA)

Introduction

The objective of this task was to explore the cleaned e-commerce sales dataset and discover meaningful patterns, trends, and business insights through data visualization and statistical analysis.

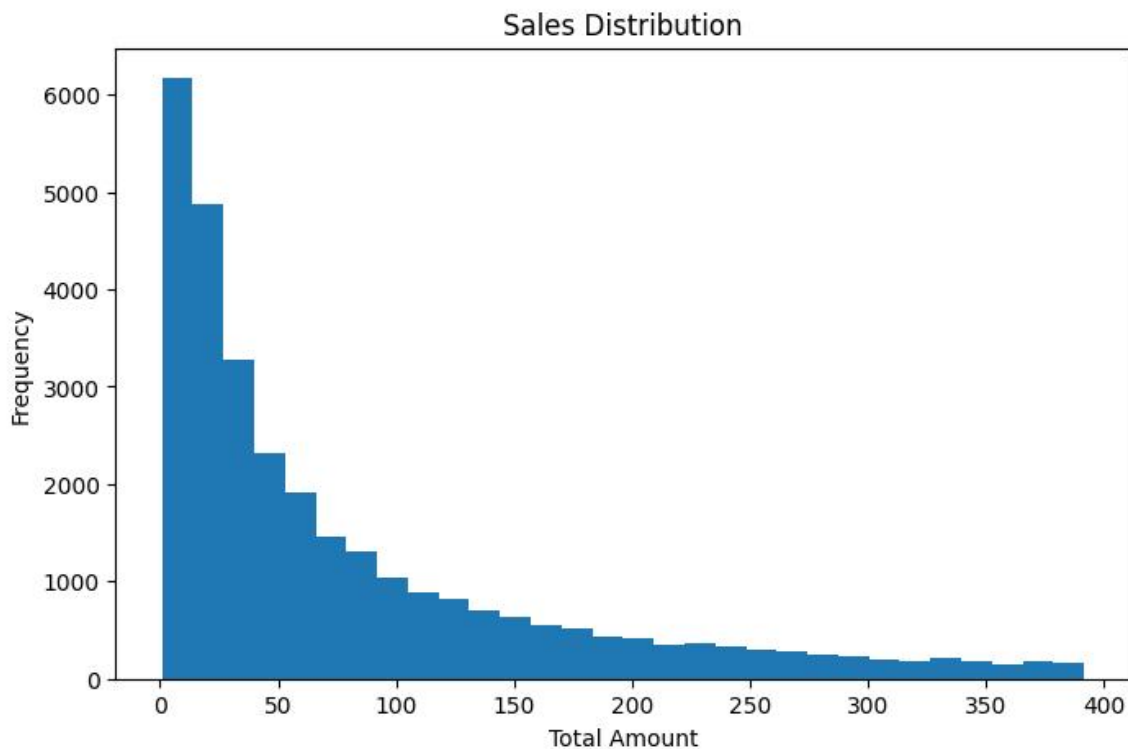
Dataset Overview

The cleaned dataset generated during Task 1 was used for exploratory analysis.

Analysis Performed

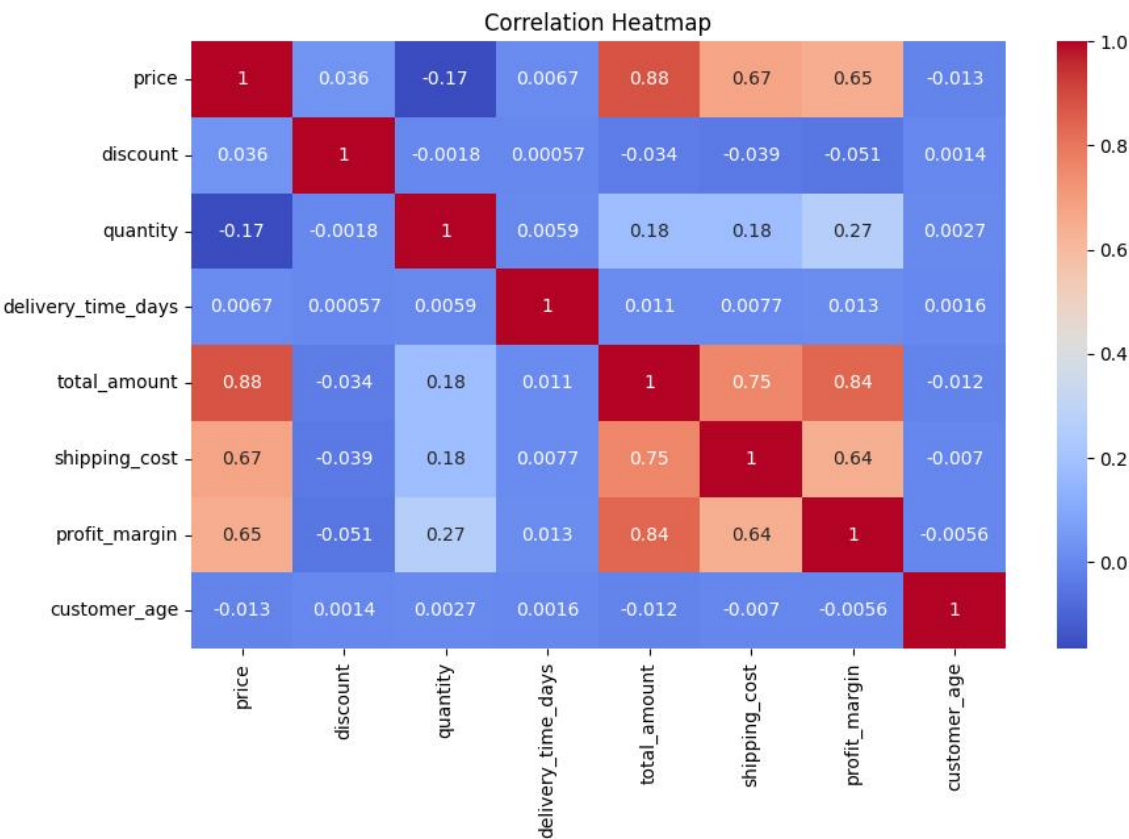
Sales Distribution Analysis

A histogram was created to analyze the distribution of total sales amounts and identify overall sales behavior.



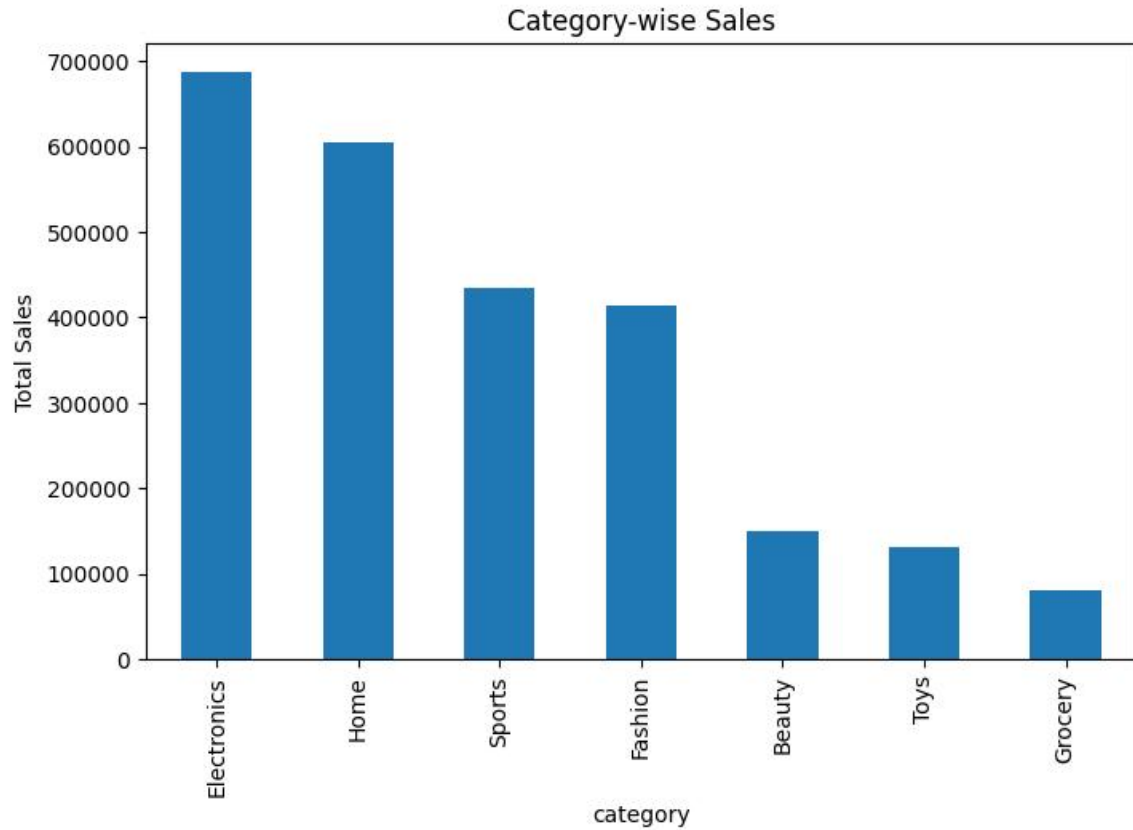
Correlation Analysis

A correlation heatmap was generated to study relationships among numerical variables such as price, quantity, shipping cost, profit margin, and total amount.



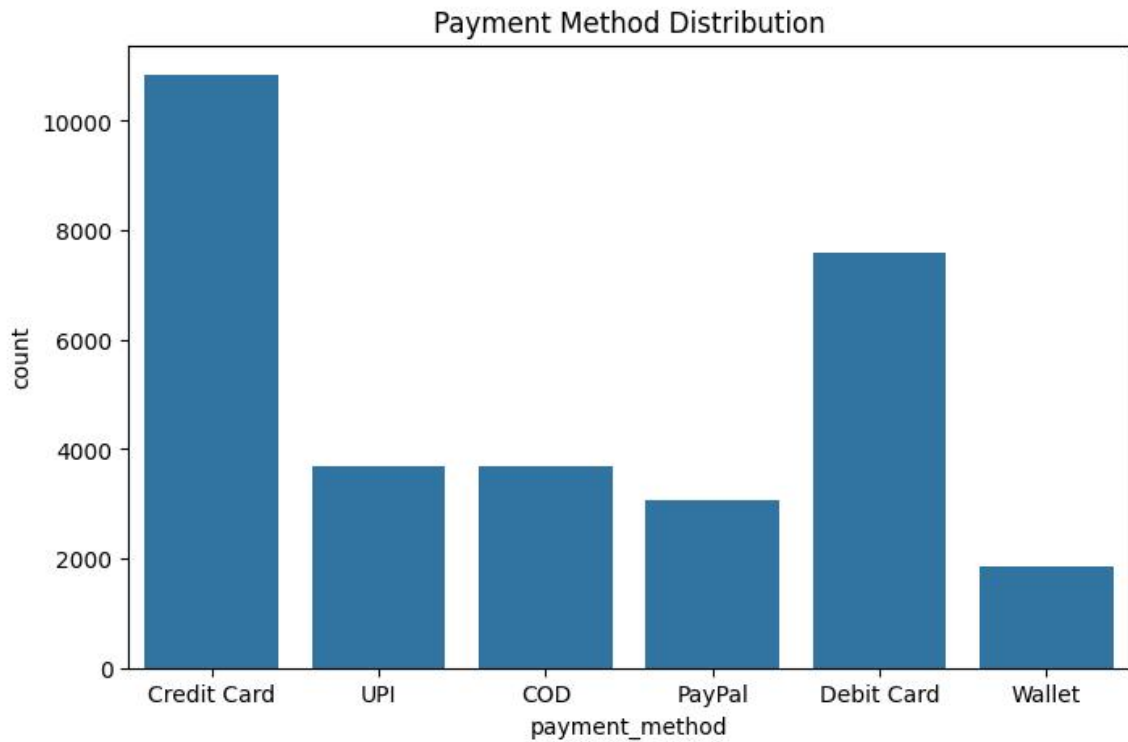
Category-wise Sales Analysis

Total sales were grouped by product category to identify the categories generating the highest revenue.



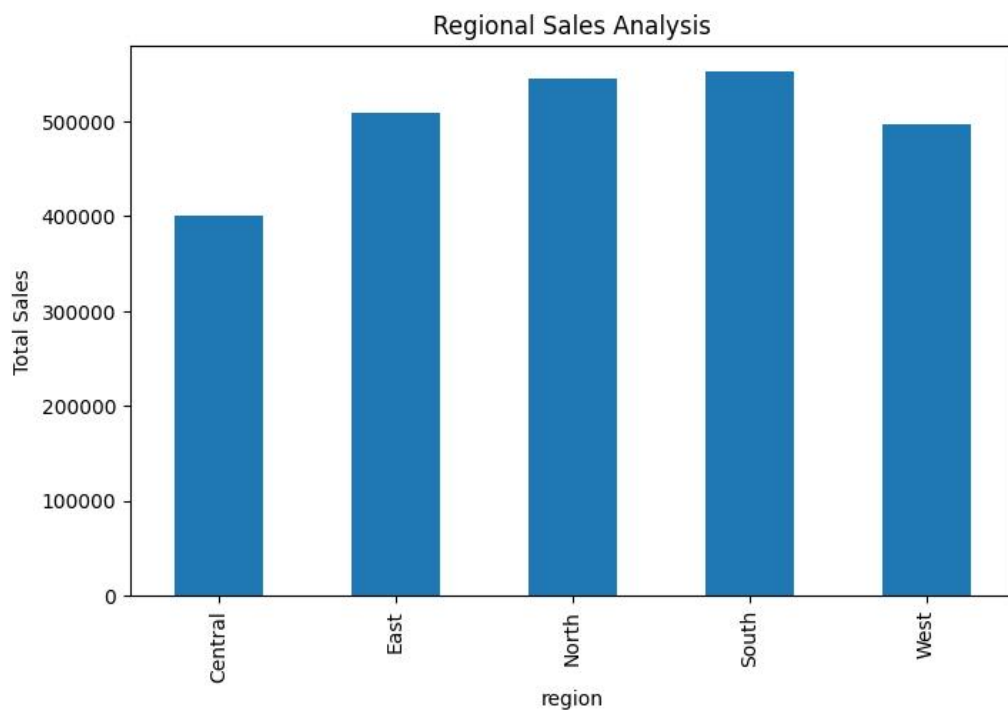
Payment Method Analysis

A count plot was used to analyze the frequency of different payment methods used by customers.



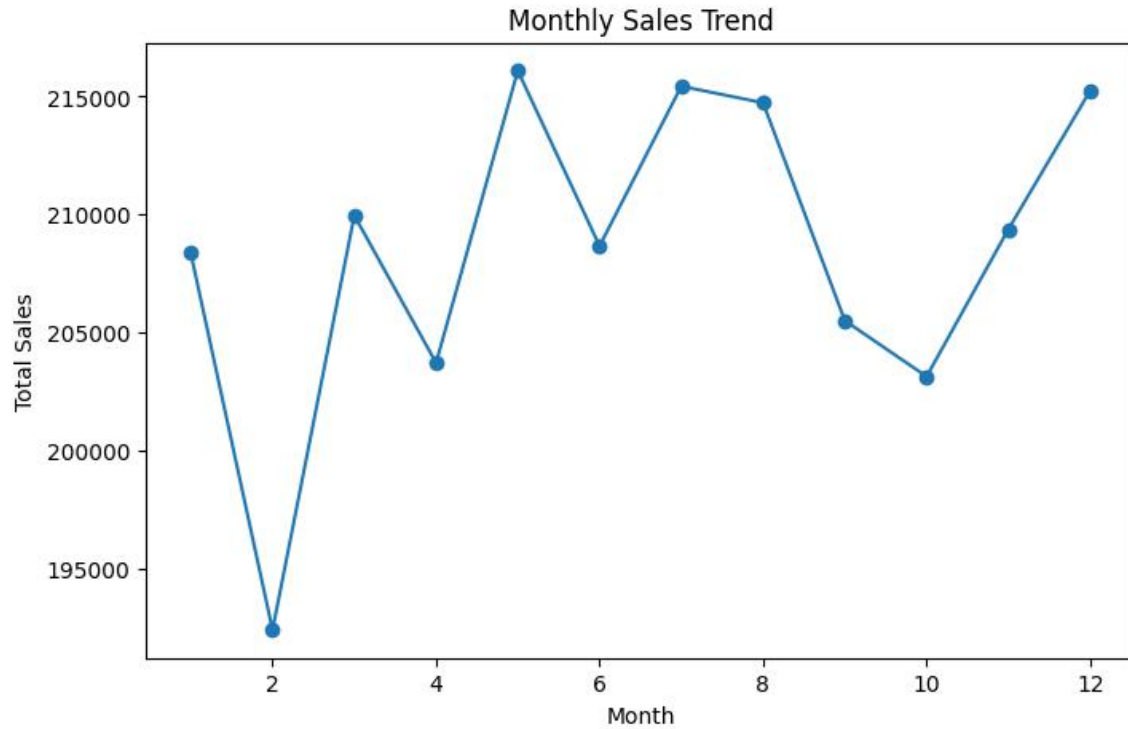
Regional Sales Analysis

Sales performance was examined across different regions using bar charts.



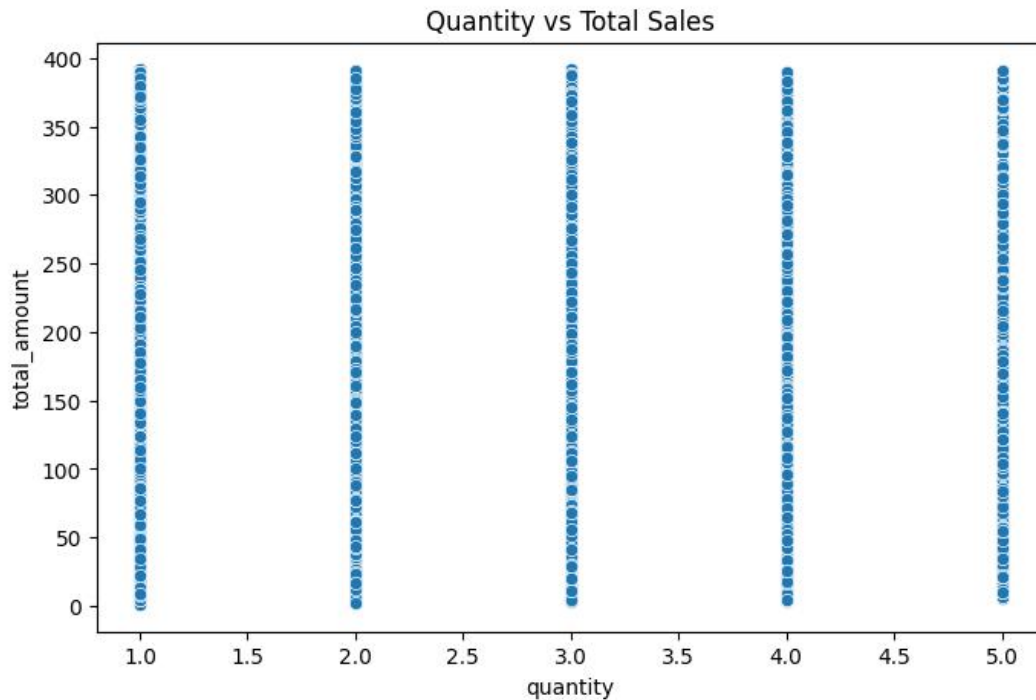
Monthly Sales Trend Analysis

The order date column was converted to datetime format and monthly sales trends were visualized to identify seasonal patterns.



Quantity vs Total Sales Analysis

A scatter plot was created to examine the relationship between product quantity and total sales amount.



Key Findings

- Product categories contributed differently to overall revenue.
- Regional sales performance varied significantly.
- Monthly sales trends indicated fluctuations in customer demand.
- Quantity sold showed a positive relationship with total sales.
- Several numerical variables displayed moderate to strong correlations.

Business Insights

- High-performing categories can be prioritized for inventory management.
- Regional trends can help optimize marketing strategies.
- Understanding payment preferences can improve customer experience.
- Sales trends can assist in forecasting future demand.

Conclusion

The exploratory analysis provided valuable insights into customer behavior, product performance, and sales trends. These findings established a strong foundation for building predictive models in the next phase of the project.